

ML Assignment 1: Polynomial Regression

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1 Introduction

The task is to predict a continuous target y for two problems using polynomial regression only. Problem 1 (var1) has six inputs $x_1, \dots, x_6$ and a target that is a polynomial of degree at most 10. Problem 2 (var2) has three inputs x_1, x_2, x_3 and a target of degree at most 20. Each problem has 1000 training rows and 1000 test rows.

The three things to decide for each problem are the degree of the polynomial, which features to use, and whether to regularize the model. All three were chosen using K-fold cross validation on the training data, with mean squared error (MSE) and R^2 as the criteria. The final models are summarised in Table 1.

Problem	Features	Degree	Model	CV MSE	CV R^2
var1	x_1-x_6	5	Lasso, $\alpha = 0.01$	0.317	0.971
var2	x_1-x_3	10	Ridge, $\alpha = 1.0$	0.259	0.994

Table 1: Final model chosen for each problem.

2 Method

2.1 Model

A polynomial of degree d in k features contains every monomial whose powers add up to at most d :

$$\hat{y} = w_0 + \sum_j w_j x_1^{a_{j1}} x_2^{a_{j2}} \cdots x_k^{a_{jk}}, \quad a_{j1} + \cdots + a_{jk} \leq d.$$

The number of such terms (excluding the constant) is $\binom{k+d}{d} - 1$, which grows quickly. For var1 with six features this is 209 terms at degree 4, 461 at degree 5 and 923 at degree 6. The model is linear in the weights w_j , so it is fit with ordinary least squares.

Before building the polynomial terms, each input is scaled to $[-1, 1]$. Without this, high powers such as x^{15} become very large or very small and the fit is numerically unstable.

2.2 K-fold cross validation

The training data is shuffled and split into $K = 10$ folds. The model is trained on nine folds and evaluated on the remaining one, and this is repeated so that every fold is used once for validation. The reported train and validation MSE and R^2 are the averages over the ten folds. The validation score estimates how the model performs on unseen data, so it is used to choose between models. A model that overfits has a low training error but a high validation error.

2.3 Selection procedure

The same three steps were applied to both problems.

1. **Degree.** With all features, every degree was evaluated: 1 to 10 for var1 and 1 to 20 for var2.

2. **Features.** Every combination of features was evaluated at every degree (63 combinations for var1, 7 for var2). The degree is varied together with the features because the best degree can change when features are removed.
3. **Regularization.** Ridge (L2 penalty) and Lasso (L1 penalty) were fit on all features for every degree, with $\alpha \in \{0.001, 0.01, 0.1, 1, 10\}$ for ridge and $\alpha \in \{0.001, 0.01, 0.1\}$ for lasso. The polynomial terms were standardised first so that the penalty treats all terms equally.

The best plain, ridge and lasso models were then compared and the one with the lowest validation MSE was trained on the full training set and used to predict on the test set.

3 Problem 1: Steam Turbine Optimization (var1)

3.1 Degree

Table 2 and Figure 1 show the degree sweep with all six features. The validation MSE falls until degree 4 and then rises again at degree 5, while the training MSE keeps falling. From degree 6 the model has 923 terms but only 900 training rows in each fold, so it fits the training data exactly (training MSE of about 10^{-26}) and the validation error becomes very large. Without regularization the best degree is therefore 4.

Degree	Terms	Train MSE	Val MSE	Train R^2	Val R^2
1	6	9.749	9.885	0.132	0.111
2	27	3.251	3.513	0.710	0.678
3	83	0.801	1.018	0.929	0.907
4	209	0.363	0.705	0.968	0.935
5	461	0.128	0.964	0.989	0.910
6	923	≈ 0	937.0	1.000	-85.6
10	8007	≈ 0	2.938	1.000	0.735

Table 2: var1: degree sweep with all features and no regularization.

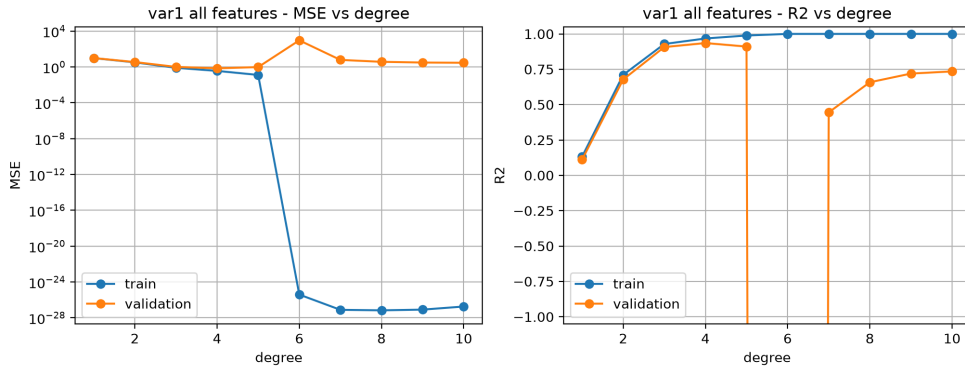


Figure 1: var1: train and validation MSE and R^2 against degree.

3.2 Features

Figure 2 shows the twelve best feature combinations out of 63, each at its best degree. Using all six features is clearly the best. The best combination with one feature removed (x_1, x_2, x_3, x_5, x_6) has a validation MSE of 1.69 compared to 0.705 with all six, and every other combination is worse. All six features were therefore kept.

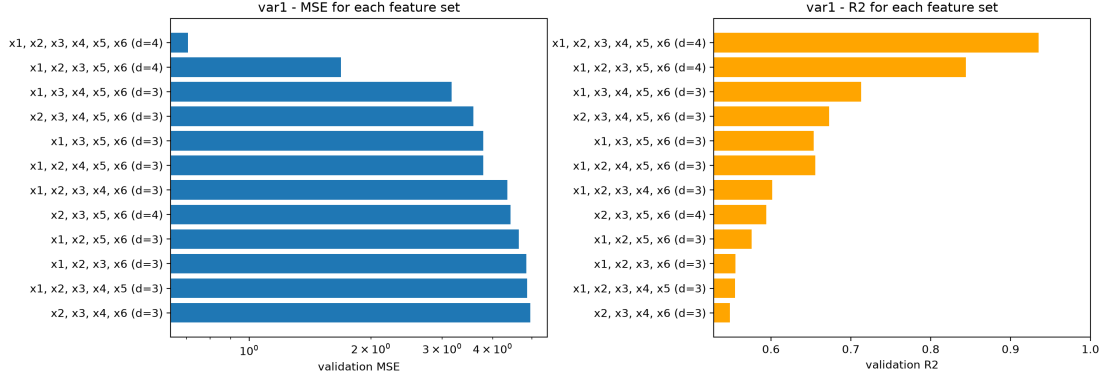


Figure 2: var1: validation MSE and R^2 of the best feature combinations.

3.3 Ridge and Lasso

The degree sweep suggests that degree 5 fits the data better than degree 4 (lower training error) but has too many terms for the amount of data. Regularization addresses exactly this. Figures 3 and 4 show the validation scores for each α .

With ridge, a larger α gives a lower error, and the best result is at degree 5 with $\alpha = 10$. With lasso, $\alpha = 0.01$ gives the lowest error at degree 5, and the error stays almost flat up to degree 10. This is because the L1 penalty sets the weights of unneeded terms to exactly zero, so the extra higher-degree terms do little harm.

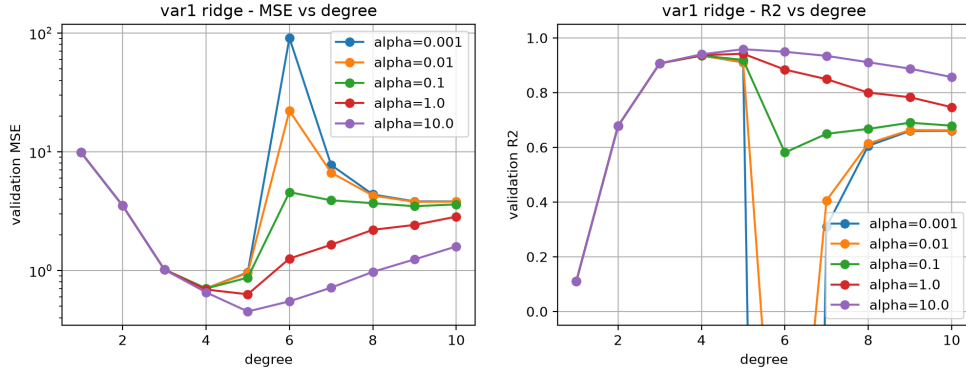


Figure 3: var1: ridge validation MSE and R^2 for each α .

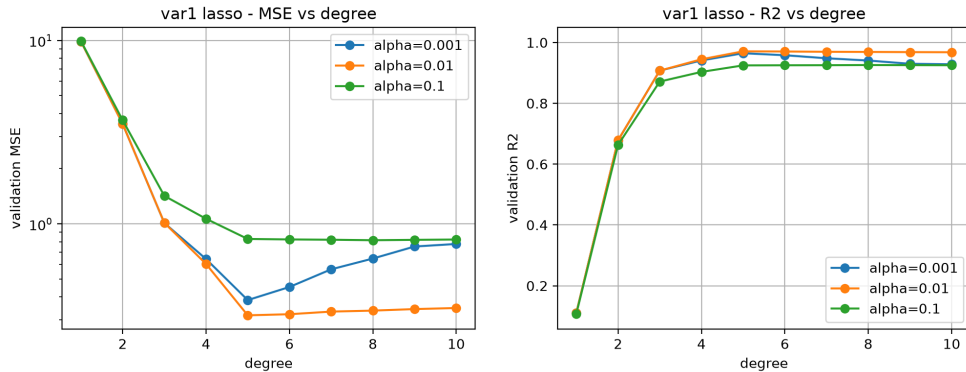


Figure 4: var1: lasso validation MSE and R^2 for each α .

3.4 Final model

Table 3 compares the best model of each type. Lasso at degree 5 reduces the validation MSE from 0.705 to 0.317, and its training MSE (0.229) is close to its validation MSE, which indicates that it

is not overfitting. The final model for var1 is a **degree 5 polynomial in all six features with Lasso**, $\alpha = 0.01$.

Model	Degree	α	Train MSE	Val MSE	Val R^2
Plain	4	—	0.363	0.705	0.935
Ridge	5	10	0.161	0.451	0.959
Lasso	5	0.01	0.229	0.317	0.971

Table 3: var1: best plain, ridge and lasso models.

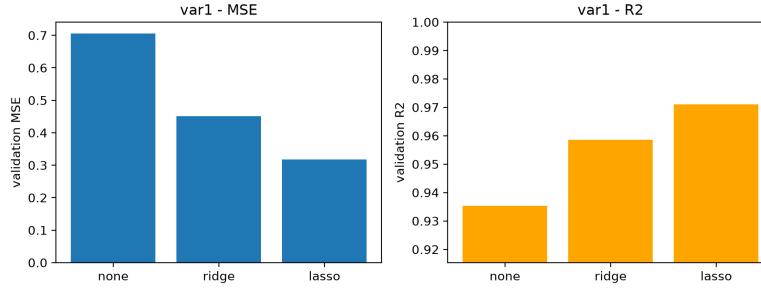


Figure 5: var1: comparison of the three models.

4 Problem 2: Thermal Reservoir Mapping (var2)

4.1 Degree

Table 4 and Figure 6 show the degree sweep with all three features. The validation MSE falls steadily up to degree 8 and stays close for degrees 9 and 10. After that it rises, slowly at first and then sharply from degree 13, while the training MSE keeps decreasing. Without regularization the best degree is 8.

Degree	Terms	Train MSE	Val MSE	Train R^2	Val R^2
2	9	19.225	19.978	0.556	0.532
4	34	3.225	3.684	0.926	0.912
6	83	0.457	0.604	0.989	0.985
8	164	0.174	0.271	0.996	0.993
10	285	0.133	0.301	0.997	0.993
12	454	0.095	1.184	0.998	0.970
14	679	0.047	82.16	0.999	-0.97
20	1770	0.019	790.5	1.000	-19.9

Table 4: var2: degree sweep with all features and no regularization (selected degrees).

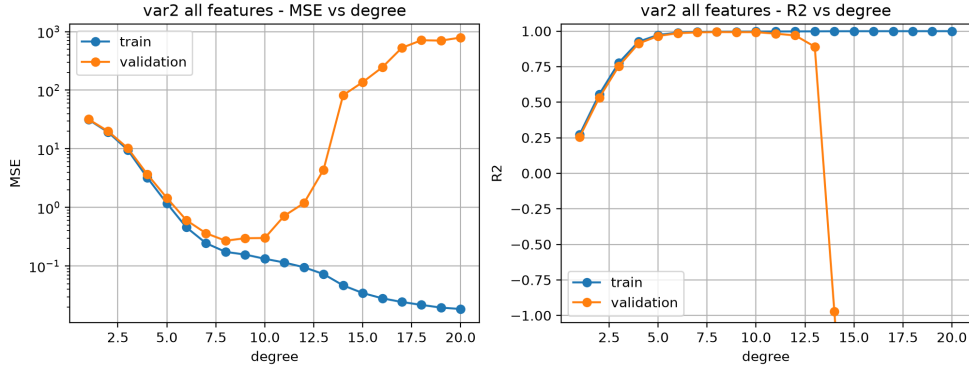


Figure 6: var2: train and validation MSE and R^2 against degree.

4.2 Features

Figure 7 shows all seven feature combinations, each at its best degree. The target depends on all three coordinates. The best pair, (x_1, x_2) , only reaches a validation R^2 of about 0.59, and no single feature gets above 0.3, compared to 0.993 with all three. All three features were therefore kept.

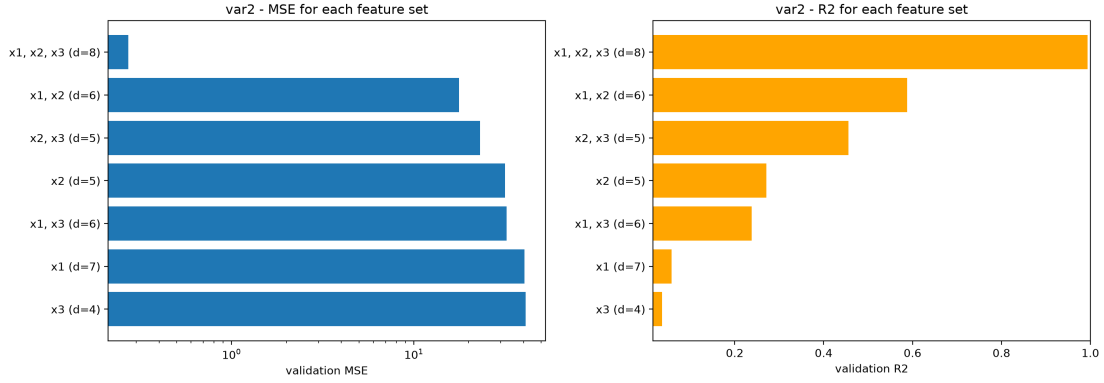


Figure 7: var2: validation MSE and R^2 of every feature combination.

4.3 Ridge and Lasso

Figures 8 and 9 show the regularized models. Both penalties remove the sharp rise in validation error at high degrees: with ridge at $\alpha \geq 1$ or lasso at $\alpha = 0.001$, the validation MSE stays near its minimum all the way to degree 20. The lowest ridge error is at degree 10 with $\alpha = 1$, and the lowest lasso error is at degree 14 with $\alpha = 0.001$.

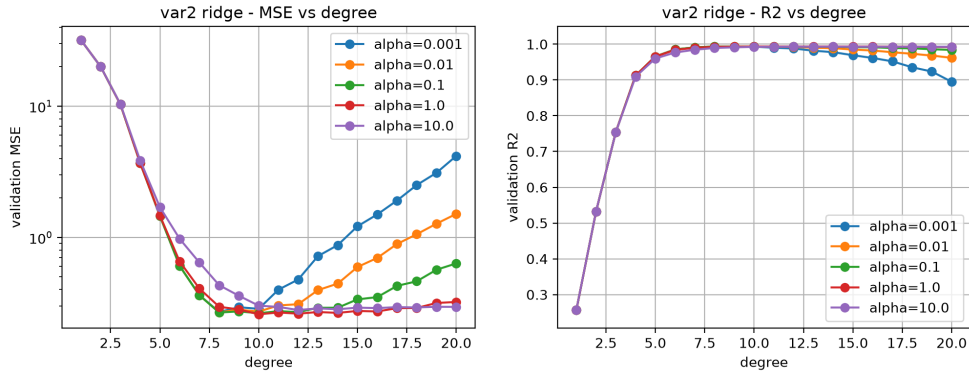


Figure 8: var2: ridge validation MSE and R^2 for each α .

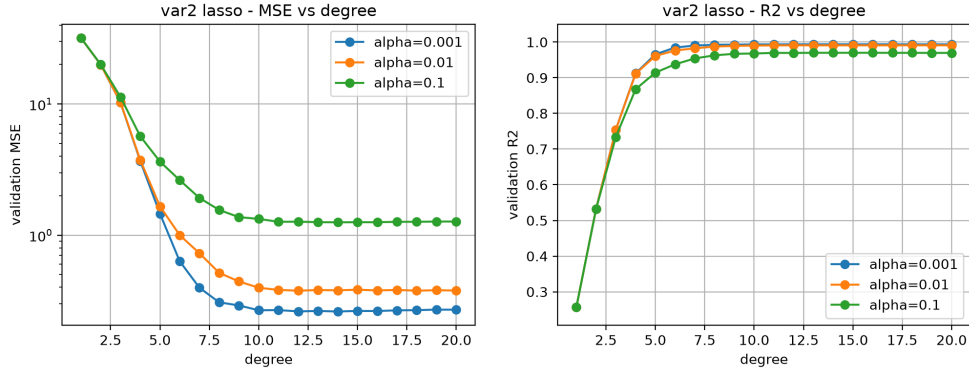


Figure 9: var2: lasso validation MSE and R^2 for each α .

4.4 Final model

Table 5 compares the best model of each type. The three are very close: the difference between the best and the worst is about 0.011 in MSE. Ridge at degree 10 has the lowest validation MSE and was selected, but the improvement over the plain degree 8 model is small. The final model for var2 is a **degree 10 polynomial in all three features with Ridge, $\alpha = 1.0$** .

Model	Degree	α	Train MSE	Val MSE	Val R^2
Plain	8	—	0.174	0.2706	0.9934
Ridge	10	1.0	0.175	0.2593	0.9937
Lasso	14	0.001	0.174	0.2615	0.9936

Table 5: var2: best plain, ridge and lasso models.

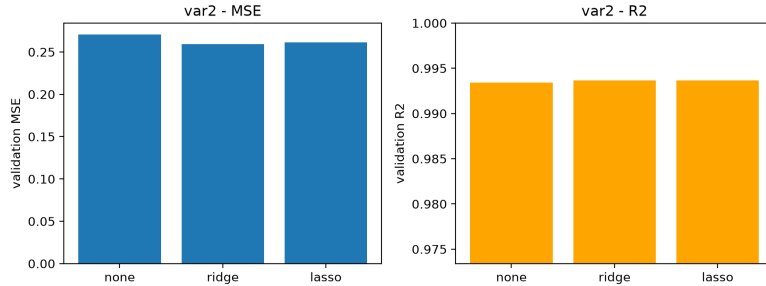


Figure 10: var2: comparison of the three models.

5 Conclusion

For both problems the degree, the features and the regularization were chosen with 10-fold cross validation using MSE and R^2 . In both cases all the given features were needed, since removing any feature increased the validation error considerably.

For var1, the unregularized model was limited to degree 4 because the number of terms grows very quickly with six features. Lasso allowed a degree 5 model to be used and reduced the validation MSE from 0.705 to 0.317 (R^2 from 0.935 to 0.971). For var2, a plain degree 8 model was already accurate, and ridge at degree 10 gave a small further improvement (validation MSE 0.259, R^2 0.994).

The reported scores are cross validation estimates on the training data. Since the final models were chosen as the best among many candidates, the error on the hidden test set may be slightly higher than these values.